Scattering Enhanced Attention Pyramid Network for Aircraft Detection in SAR Images

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***Abstract*— Aircraft detection in synthetic aperture radar (SAR) images is a challenging task because of the discreteness, variability, and interference of aircraft scattering features. This article proposes a new hybrid approach of scattering information enhancement (SIE) and an attention pyramid network (APN). It first extracts strong scattering points (SSPs) of aircraft via an adapted Harris–Laplace detector. These SSPs are then clustered into candidate scattering regions by density-based spatial clustering of applications with noise (DBSCAN) and are then modeled with a Gaussian mixture model (GMM). Target scattering clusters are discriminated from background clutter by measuring the Kullback–Leibler divergence (KLD) to the known target templates. These target scattering clusters are enhanced in the preprocessing stage. All the SIE-preprocessed images are then fed into the APN for training and testing. It is composed of the multiscale feature pyramid network (FPN) and the modified convolutional block attention module (CBAM) to cope with the discreteness and variability of aircraft. In addition, focal loss (FL) is adopted to deal with the issue of unbalanced sample distribution and the interference from hard samples. Experiments conducted on the Gaofen-3 and TerraSAR-X data sets demonstrate the effectiveness of the proposed method with an average precision (AP) of 83.25%.**

***Index Terms*— Attention pyramid network (APN), convolutional block attention module (CBAM), focal loss (FL), Gaussian mixture model (GMM), scattering information enhancement (SIE), synthetic aperture radar (SAR) aircraft detection.**

NOMENCLATURE

AD	Airport detection.
AIO	Adaptive identification operator.
AIC	Akaike's information criterion.
AM	Attention module.
APN	Attention pyramid network.
CBAM	Convolutional block attention module.
CE	Cross entropy.
DBSCAN	Density-based spatial clustering of applications with noise.
EM	Expectation–maximization.
FL	Focal loss.
FPN	Feature pyramid network.
GMM	Gaussian mixture model.
KLD	Kullback–Leibler divergence.
SE	Squeeze and excitation.
SIE	Scattering information enhancement.
SSP	Strong scattering point.

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I. INTRODUCTION

SYNTHETIC aperture radar (SAR) is an active microwave imaging sensor with an all-day and all-weather imaging observation capability [1], [2]. It has been widely used in various fields for its unique characteristics since its birth in the 1950s. With the development of the SAR imaging technique and improvement of the image resolution, typical target monitoring with SAR images has attracted much attention [3], [4]. As a fundamental task, target detection is used to obtain the accurate position of a target with both a high detection rate and a low false alarm rate. Aircraft detection in high-resolution SAR images is one of the major tasks in SAR target detection and has great applications in various fields.

Traditional SAR target detection algorithms include three major categories: single-feature-based, multifeature-based, and expert-system-based [5], in order of their increasing performance but also their algorithm complexity. The single-feature-based algorithms use a single feature to describe the target in SAR images, typically the pixel intensity. The constant false alarm rate (CFAR) based on clutter statistics and threshold extraction is a typical feature extraction method based on intensity features [6]. Various CFAR-derived algorithms have been proposed, such as cell-averaging constant false alarm rate (CA-CFAR) [7], the smallest of cell-averaging constant false alarm rate (SOCA-CFAR) [8], the greatest of cell-averaging constant false alarm rate (GOCA-CFAR) [9], ordered statistic constant false alarm rate (OS-CFAR) [10], and variability index constant false alarm rate (VI-CFAR) [11]. The single-feature-based approach is the basic building block for the other two categories. The multifeature-based algorithms fuse two or more features. Besides intensity, other features that can be inferred from SAR images include the target contour, target size, texture, and scattering points of the target [12]–[15]. The expert-system-based method is the most sophisticated and further utilizes some expert knowledge

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about the imaged scene, clutter, and/or targets. Template- and model-based algorithms are typical expert-system-based methods. Template-based methods measure the similarity between the target and the template by features, such as size, direction, and image elements. Their performance relies on the completeness of the template database. Model-based methods iteratively match the target with simulated SAR images and have very high requirements for both the speed and accuracy of the electromagnetic simulator [16].

With the rapid development of artificial intelligence, deep learning has achieved great success in many applications [17], including target detection in the computer vision area. These methods mainly deal with optical images. Girshick *et al.* [18] proposed the region-based convolutional neural network (R-CNN) for object detection through four steps, i.e., generating candidate regions, extracting features with the CNN, image classification, and nonmaximum suppression for predicted bounding boxes. He *et al.* [19] introduced a spatial pyramid-pooling layer and proposed spatial pyramid-pooling-net (SPP-net), which significantly shortened the training time. Fast R-CNN was proposed on this basis and samples feature graphs of different sizes into a fixed size, using only one scale for grid generation and pooling to further improve the computing speed [20]. Ren *et al.* [21] introduced the region proposal network (RPN) into fast R-CNN and proposed a faster R-CNN that sets different scales of anchors by certain rules to extract region proposals in the convolution feature layer. In order to enhance the translation sensitivity of the network and improve the positioning accuracy of the object, region-based fully convolutional network (R-FCN) was proposed, which shares the computation of the regions of interest [22]. The FPN was further proposed, which uses multiscale features and a top–down structure for object detection, effectively improving the detection performance for small targets [23]. Many other methods have been proposed for object detection (see [24]–[29]).

Deep learning has also been applied in SAR image interpretation [30]–[37], including aircraft detection, which will be discussed later in Section II. However, there are still challenges to be addressed. For example, the deep learning algorithm may not be suitable for the unique imaging mechanism of SAR and the complexity of the aircraft structure. The imaging results for the same target may be different under different imaging conditions, which is fatal to algorithms relying heavily on training data such as deep learning. Target scattering is one of the major forms of features in SAR images. It has not been fully understood and explored in deep learning-based approaches. This article aims to combine SIE with the APN deep learning approach to deal with the specific task of aircraft detection in SAR images. The proposed approach is validated on a data set constructed and manually labeled from the Gaofen-3 and TerraSAR-X images.

The rest of this article is organized as follows. Section II briefly reviews related work and summarizes the key challenges. Section III describes the proposed SIE-APN algorithm in detail. Experimental results as well as performance evaluation are presented in Section IV. Section V summarizes this article and provides an outlook on future work.

II. Related Work

A. Nondeep Learning Methods

Traditional methods are mainly based on manually designed features from the target structure and scattering features. Target structure features include the target contour, component shape, and structure distribution. An aircraft detection and recognition method based on structure characteristics in SAR images is proposed, which is based on the aircraft skeleton structure and key size information [38]. The scattering features used for aircraft detection in single-polarization SAR images can be mainly divided into three categories: geometric features, statistical features, and texture features. Due to the unique imaging mechanism of SAR, aircraft in SAR images is usually composed of a series of SSPs. Therefore, geometric features, such as scattering points or the edge density, can be considered for aircraft interpretation. The scattering characteristics of civil aircraft in the TerraSAR-X data are analyzed and described by a convex point vector, which realizes the detection and recognition of aircraft [39]. An edge detection algorithm based on the Canny operator is introduced to extract candidate slices of aircraft in [40]. Statistical feature-based methods generally extract features from the data first with ground truth and then take these features to establish a database or an iterative model, based on which the target identification can be realized. The GMMs are introduced to model aircraft slices and a similarity measurement algorithm based on KLD is utilized for aircraft slices identification. Similarly, GMM is utilized for modeling aircraft distribution, and template matching is used to improve the accuracy and efficiency of the recognition algorithm [41]. A multicomponent model based on mixed statistical distribution is established to distinguish the aircraft from the background [42]. The common texture features for aircraft interpretation in SAR images include direction gradient distribution and visual significance. An aircraft detection method based on the direction local gradient distribution is proposed, which realizes the complete extraction and combination of aircraft by applying interactive constraints between the adjacent pixels based on the calculated probability [43].

B. Deep Learning-Based Methods

Deep learning has demonstrated a powerful representation ability in the field of computer vision, which can achieve satisfactory results using CNNs instead of manually designed features. CNNs have been adopted in various stages for aircraft interpretation, including ROI detection, false alarm discrimination, and object recognition. An aircraft detection algorithm in SAR images based on data enhancement and a CNN is proposed in [44]. First, the candidate slices are selected by a sliding window, and then, the ROIs are identified by a convolutional neural network. Considering that the target contour is important *a priori* information for target recognition, an aircraft reconstruction method using the shape *a priori* to accurately generate the contour is proposed in [45]. The method is divided into two stages: first, an *a priori* target shape is obtained by using the generative deep learning modeling, and then, a rough-fine target attitude estimation method is adopted to obtain the precise target

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contour. At the present stage, most of the aircraft detection algorithms tend to use an end-to-end detection framework that inputs the original image to the network and outputs the detection results from the network directly. He *et al.* [46] built a component-based multilayer parallel network for airplane detection in SAR imagery, and the experiment conducted on TerraSAR-X aircraft cemetery data proved that this method was feasible. In order to describe the target position more accurately, An *et al.* [47] adopted rotating rectangular boxes and FPN to improve the detection performance for small targets. Zhang *et al.* [48] put forward a cascade three-view network that is based on a faster R-CNN structure for aircraft detection in SAR images. Considering that the aircraft in the SAR image has the characteristics of discreteness, a pyramid attention dilated network is proposed, which is composed of two key components: multibranch dilated convolution module and CBAM [49].

C. Major Challenges in SAR Aircraft Detection

The major challenges in problems of aircraft detection methods in SAR images are summarized as follows.

1) Complex scattering characteristics of aircraft are as follows.
   a) *Discreteness:* Due to the smoothness of the aircraft surface, its imaging result tends to appear as discrete scattering points in high-resolution SAR images, which means that an aircraft is easily divided into many small pieces. In this case, it is difficult to detect the object completely.
   item *Variability:* Due to the aircraft's complex structure, different components of it have different scattering mechanisms. In the process of SAR imaging, with the change of scattering conditions, the imaging results also change to different degrees.
   item *Interference:* A large number of strong background scattering points are usually distributed around the aircraft, which may be easily confused with the components of the target, making accurate detection difficult [50].
   b) *Infeasibility of Transfer Learning From Optical Networks:*
   The existing deep learning-based detection algorithms in SAR imagery are mainly based on the fine-tuning for typical computer vision networks, and they do not combine the SAR imaging mechanism as well as the scattering characteristics of targets. The imaging mechanisms of SAR and optical satellites are different. SAR is sensitive to imaging conditions and observation angles. Therefore, the algorithm based on an optical image pretrained network cannot be directly transferred to SAR images.
2) item *Redundancy of Feature Extraction Results in deep CNN:* Different channels and spatial dimensions of the feature maps that are extracted from the deep CNN have different importance. In particular, when combining target scattering information and utilizing the multiscale structure in the detection algorithm, the feature extraction results aggregate context information in space and quantities of information from different receptive fields. In this case, the network is prone to develop feature redundancy, which is unfavorable for model generalization for different sizes of aircraft.

D. Proposed Approach

Intending to solve the abovementioned challenges, we propose an aircraft detection algorithm in SAR imagery based on an APN with SIE and apply it to detect aircraft in both complex backgrounds and target dense arrangement situations.

The contributions of this approach are in four aspects.

1) An AD method based on AIOs [40] is proposed to extract the airport mask precisely, which can reduce the influence of complex background, avoiding a large number of false alarms as well as improving the algorithm efficiency.
2) A scattering enhancement strategy is designed for enhancing the brightness information of those areas predicted to contain aircraft in the SAR images by extracting, modeling, and identifying the target scattering information, which is beneficial to cope with the variability caused by the unique scattering mechanism. Compared with the original imagery, the enhanced imagery highlights the aircraft characteristics as well as weakening the background clutter, thus providing effective prior information for the subsequent network.
3) An APN for aircraft detection is designed, which is composed of an FPN structure and an AM. The FPN is introduced to utilize multiscale information for improving the detection performance and considers both the high-resolution characteristics of the shallow layers and the high-semantic characteristics of the deep layers. The AM is proposed to improve the target features extraction ability for different scales of aircraft and is mainly composed of a CBAM [51]. Besides, FL is utilized as a classification loss function to deal with the influence of unbalanced distribution of positive and negative samples as well as hard samples [52].
4) Comprehensive ablation experiments are conducted in the Gaofen-3 and TerraSAR-X satellite data sets to reveal the effect of each improvement as well as to demonstrate the robustness of the proposed algorithm on detection results.

III. Methodology

A. Overview

The framework of the proposed SIE-APN method is shown in Fig. 1, which can be divided into five stages: AD, scattering information extraction, scattering region discrimination, deep characteristic extraction, and prediction.

1) *AD Stage:* An AD algorithm based on morphological processing and AIOs is proposed, which can accurately segment the airport area by identifying the acquired connected domain. The AD algorithm can reduce the influence of complex background clutter as well as suppressing a lot of false alarms.

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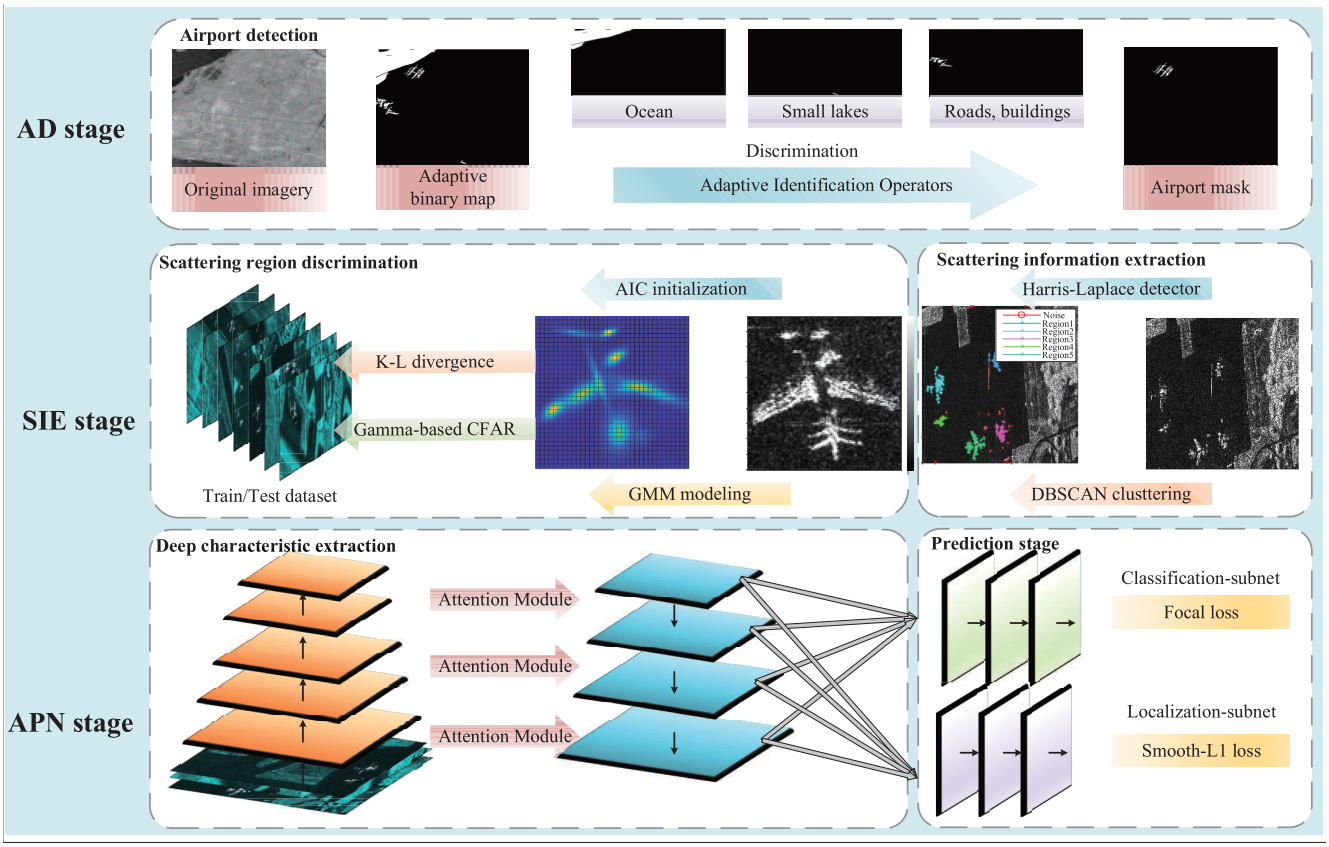

Fig. 1. Framework of the proposed SIE-APN method.

2) *Scattering Information Extraction Stage:* Considering the unique scattering mechanism of aircraft in SAR images, various scattering results can be described essentially as the combination of SSPs. Therefore, SSPs that can be extracted by the Harris–Laplace detector with scale invariance and rotation invariance are selected as the basis for modeling the scattering information of aircraft in this article. Especially, the DBSCAN algorithm is introduced to cluster the extracted SSPs information to generate scattering region proposals because the distribution density of different aircraft is small, while that of the same aircraft components is large in an SAR imagery scene. This satisfies the prerequisite of density clustering; the clustering structure can be determined by the degree of density of the sample distribution.
3) *Scattering Region Discrimination Stage:* A GMM is used to model the pixel intensity distribution of the SSPs and their corresponding scattering regions, whose pixel intensity represents the frequency of occurrence. The GMM requires the number of clusters components to be specified before being fitted to the data, but it might be difficult to obtain the appropriate number of components in practice. Thus, AIC is introduced to explore the data and make an initial guess at the number of components. With the initialization of the AIC, the GMM is optimized by the EM algorithm. In order to identify the fitted scattering regions, KLD is introduced to measure the similarity between the reference GMM and the candidate GMM. It is worth mentioning that this step is essentially equivalent to a two-classification process between the scattering regions including aircraft and background targets, which are quite distinctive from each other. Therefore, KLD can be used as a measurement index to obtain a strong identification effect. The scattering regions predicted to contain aircraft will be enhanced with brightness information through gamma distribution-based CFAR. The essence of SIE is to highlight the aircraft features and weaken the influence of background targets by utilizing prior knowledge and modeling the scattering information of the aircraft, which has the benefit of mitigating the variability and providing more effective information for the neural network.
4) *Deep Characteristic Extraction Stage:* An APN structure is proposed by introducing multiscale information and an AM, which is composed of FPN and AM. The FPN module extracts the deep features based on the combination of high-resolution information in shallow layers and high-semantic information in deep layers, which has the benefit of improving the detection performance. The AM reduces redundant feature extraction by inferring attention mapping from different dimensions and enables the network to pay more attention to the target itself.

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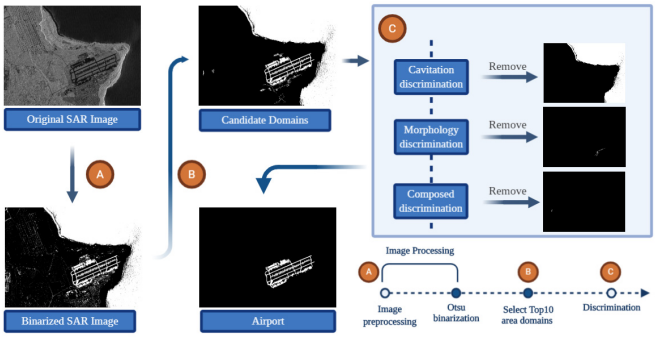

Fig. 2. Flowchart of the proposed AD method.

5) *Prediction Stage:* FL is introduced in the proposed architecture to replace the standard CE function to calculate the classification loss, which not only solves the problem of an unbalanced distribution of positive and negative samples but also achieves the appropriate distinction between simple samples and hard samples, improving the detection performance by ensuring the running speed. FL is achieved by introducing a balance factor to reshape the standard CE loss as well as reducing the loss assigned to negative samples, to cope with the unbalanced distribution of positive samples and negative samples. In addition, by changing the parameter to reduce the loss of high confidence samples, the model pays more attention to the hard samples.

In Sections III-B–III-D, the proposed SIE-APN is introduced from the following three aspects: AD stage, SIE stage composed of scattering information extraction and scattering region discrimination, and finally the APN stage with deep characteristic extraction and prediction.

B. Airport Detection

Based on the fact that aircrafts are only located at airports, it is necessary to extract airports from large scenes of SAR images. This will dramatically reduce the computing cost of the detection algorithm. Furthermore, accurate extraction of the airport can reduce the false alarm rate in an end-to-end automatic detection process. In an SAR image with a large scene, the airport has obvious contrast information and unique morphological features. Airports are usually distributed as dark areas in SAR images due to the existence of their runways. Compared with the other connected domains extracted, the airport area always has clear linear information and hole information, so the morphology features of the airport are relatively inerratic. In addition, the area of the airport and other size information can also be used as important prior information to identify the airport. Based on the abovementioned analysis, an AD algorithm based on Otsu and AIOs is proposed, whose flowchart is shown in Fig. 2.

Based on the visual attention mechanism, AD in SAR images with large scenes starts from dark region extraction. First, a filtering algorithm is utilized in order to reduce the influence of clutter on airport extraction, and then, a global threshold is determined by the Otsu algorithm, which is used to binarize the image. Because speckle noise is reflected in the granular form in SAR images, the presence of speckle noise does not significantly affect the contrast information of airports in SAR images. Therefore, the corresponding threshold of the image gray histogram intervalley point is used as the threshold for Otsu so that the gray level of the image is relatively equitably quantified to the two gray levels of 0 and 1. This binarization method has achieved stable results in the experimental data. In addition, considering that speckle noise is multiplicative noise, the average filtering algorithm is adopted to smooth the image and reduce the influence of discrete noise in the image on morphological processing. After the binarized images are processed by morphology including dilation and erosion processing, the following identification steps are carried out for the top ten connected domains determined by the area prior information, and the accurate airport mask can be finally extracted. The proposed AIOs that are applied to discriminate the connected domains are given in the following.

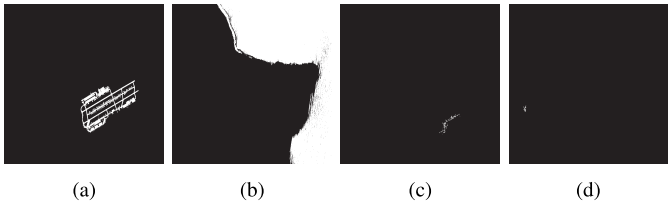

Fig. 3. Main types of connected domains in SAR imagery. (a) Connected domain of the airport. (b) Ocean. (c) Connected domain composed of intricate roads. (d) Small lakes.

1) *Cavitation Discrimination:* The ratio of the difference between the connected domain area before and after hole-filling S_{filled} and the original connected domain area S_{con}

$$k_1 = \frac{S_{\text{filled}} - S_{\text{con}}}{S_{\text{con}}}. \tag{1}$$

2) *Morphology Discrimination:* The ratio of a connected domain area S_{con} to its adjacent rectangles area S_{bbox}

$$k_2 = \frac{S_{\text{con}}}{S_{\text{bbox}}}. \tag{2}$$

3) *Area Discrimination:* The ratio of the connected domain area and the maximum connected domain area in the image

$$k_3 = \frac{S_{\text{con}}}{S_{\text{maxcon}}}. \tag{3}$$

The connected domains with a large area in SAR imagery can be mainly divided into four types, as shown in Fig. 3. Fig. 3(a) shows the connected domain of the airport. Affected by buildings in the airport, the connected domain has a large area of holes, but its shape is relatively regular, which is usually square or triangle. Fig. 3(b) shows the ocean. In this type of connected domains, there are fewer holes, larger area, and unfixed shape, which exists in the edges of the image. Fig. 3(c) shows the connected domain composed of intricate roads and so on. There are also many holes in the connected area. Due to the diverse distribution of roads, the shape is irregular.

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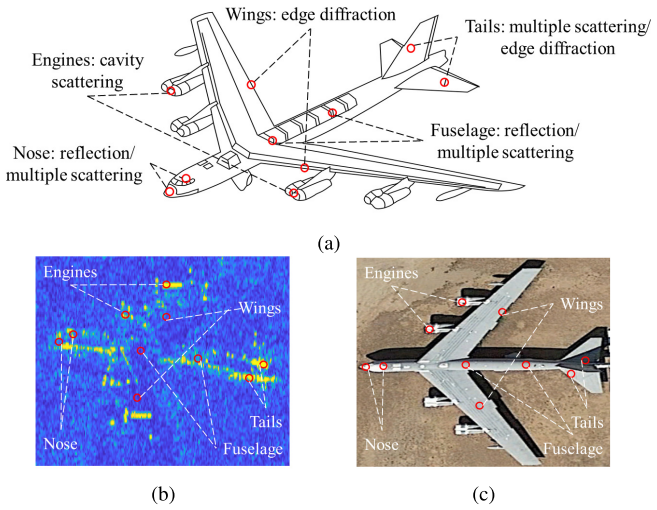

Fig. 4. Scattering mechanism of each aircraft component. (a) Structure of aircraft. (b) SAR image of aircraft. (c) Optical image of aircraft.

Fig. 3(d) shows a small lake in the large-scene SAR image, with fewer holes in the connected domain, but the area is much smaller than other connected domains. Therefore, as shown in Fig. 2, these connected regions can be discriminated by the cavitation discrimination operator, morphology discrimination operator, and composed area discrimination operator in turn.

C. Scattering Information Enhancement

Target scattering features are the basic information for the understanding and application of SAR images. In single-polarization high-resolution SAR images, because the size of the aircraft components is much larger than the resolution unit, the scattering results of the aircraft are mainly contributed by its components, and different components have different scattering mechanisms [50], as shown in Fig. 4.

Experiments show that the extraction of SSPs is stable and does not require prior knowledge. In addition, the distribution of SSPs of aircraft in SAR images is generally as circles or ellipses. Therefore, an SIE method based on the extraction of SSPs and the GMM is proposed to enhance the scattering information of areas that may contain aircraft in a scene image in advance, which will provide efficient prior information for the subsequent APN. The SIE stage is composed of the scattering information extraction stage and the scattering region discrimination stage, and the detailed descriptions are shown in the following.

1) Scattering Information Extraction Stage: The SSP is the peak value of the local scattering region, which has strong gradient information, and it changes little with the external conditions and geometric structure. Therefore, the SSPs are selected as the basis of the scattering information. The Harris–Laplace detector is utilized to extract SSPs because of its rotation invariance and scale invariance. First, the Harris detector is used for corner detection in multiscale space. Then, by calculating the Laplace of Gaussian (LoG) of the corner points, the characteristic scale values of the corner points are automatically searched for. The representation of the second moment in multiscale is given in the following, where $g(\sigma_I)$ represents the Gaussian convolution kernel with scale σ_I and x is the pixel position. σ_I is known as the integral scale and σ_D is the differential scale or local scale. L_x is the image after Gaussian smoothing and $\otimes$ denotes the convolution

$$M = {\sigma_D}^2 g(\sigma_I) \otimes \begin{bmatrix} L_x^2(x,\sigma_D) & L_x L_y(x,\sigma_D) \\ L_x L_y(x,\sigma_D) & L_y^2(x,\sigma_D) \end{bmatrix}. \quad (4)$$

Aircrafts in SAR images appear as discrete SSPs. However, these SSPs of the same target are still closer than those from different targets or noises in the same scene. DBSCAN is a good solution to cluster those points, and it is used here to cluster the extracted SSPs for generating the initial candidate scattering region; the clusters will be expanded continuously based on the connectable SSPs to obtain the final results. DBSCAN describes the degree of compactness of the sample distribution based on a set of neighborhood parameters $(\epsilon, \text{Minpets})$. For a given data set $D = \{x_1, x_2, \ldots, x_m\}$, these concepts are defined as follows.

1) *ϵ-Neighborhood:* For $x_j \in D$, the ϵ-neighborhood of x_j contains the samples whose distances between x_j are less than ϵ, that is, $N_\epsilon(x_j) = \{x_j \in D \mid \text{dist}(x_i, x_j) \le \epsilon\}$.
2) *Core Object:* If the ϵ-neighborhood of x_j contains Minpets samples at least, that is, $|N_\varepsilon(x_j)| >$ Minpets, x_j can be recognized as a core object.

Based on the abovementioned concepts, the clusters generated by DBSCAN satisfy the connectivity and maximality that involve all density-connected SSPs. Meanwhile, the SSP x_j, whose distance between core objects is greater than ϵ or $|N_\varepsilon(x_j)| \le$ Minpets, is recognized as noise or an anomalous sample and will be eliminated. The pseudocode of the proposed scattering information extraction algorithm is shown in Algorithm 1. n is determined by the size of the image and the initial scale-space variance σ. The larger the image size and σ, the larger n. The scattering information extraction result of an SAR imagery scene is shown in Fig. 5.

2) Scattering Region Discrimination Stage: Based on the extracted scattering region in the scattering information extraction stage, a GMM is introduced to model the scattering region represented for the aircraft components, which are distributed as circles or ellipses. Besides, compared with other models such as k-means and learning vector quantization which uses prototype vectors for clustering, GMM uses a probability model to express the clustering prototype, which is considered to be more reasonable and practical for aircraft in SAR imagery. The 2-D coordinates of the image represent the location of the sample, and the corresponding pixel intensity represents the probability of occurrence. The representation of the GMM is shown in the following:

$$p(x) = \sum_{i=1}^{k} \alpha_i \cdot p(x \mid \mu_i, \Sigma_i) \quad (5)$$

where k represents the number of mixture models, μ_i and Σ_i are, respectively, the mean vector and covariance matrix, and α_i is the corresponding mixture coefficient. An EM algorithm is used to solve the GMM; in each iteration, the posterior probability of each sample belonging to each Gaussian component is first calculated according to the current parameters (E step),

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Algorithm 1 Scattering Information Extraction

Input:

Sample set $S = \{s_1, s_2, \ldots, s_p\}$;
Neighborhood parameter $(\epsilon, Minpets)$;
Hyper-parameter $k = 1.4, s = 0.7$;
High threshold Th_1;
Low threshold Th_2;

Output:

Generated clusters $C = \{C_1, C_2, \ldots, C_k\}$;

Initialize corner object set $D = \emptyset$;
Determine a scale space $\sigma_1, \ldots, \sigma_n = \sigma_0, \ldots, k^n\sigma_0$;
Calculate $\Theta = \det(\mu(x, \sigma_n) - \alpha \text{trace}^2(\mu(x, \sigma_n)) > Th_1)$;
for $x \in \Theta$ **do**
  if $\mathrm{F}(x, \sigma_n) = \sigma^2 |L_{xx}(x, \sigma_n) + L_{yy}(x, \sigma_n)| \geq Th_2$ **then**
    if $\mathrm{F}(x, \sigma_n) > \mathrm{F}(x, \sigma_l), l \in \{n-1, n+1\}$ **then**
      Add x into the corner object set $D = D \cup \{x\}$;
    end if
  end if
end for
Initialize core object set: $\Omega \in \emptyset$;
for $j = 1, 2, \ldots, m$ **do**
  Determine the ϵ-neighborhood $N_\epsilon(x_j)$ of x_j;
  if $|N_\epsilon(x_j)| \geq Minpets$ **then**
    Add x_j into the core object set: $\Omega = \Omega \cup \{x_j\}$;
  end if
end for
Initialize number of clusters: $k = 0$;
Initialize unvisited sample set: $\Gamma = D$;
while $\Omega \neq \emptyset$ **do**
  Record current unvisited sample set: $\Gamma_{old} = \Gamma$;
  Select a core object $o \in \Omega$ randomly and initialize queen $Q = < o >$;
  $\Gamma = \Gamma \setminus \{o\}$;
  while $Q \neq \emptyset$ **do**
    Select the first sample q of Q;
    if $|N_\epsilon(q)| \geq Minpets$ **then**
      Set $\Delta = N_\epsilon(q) \cap \Gamma$;
      Add samples of Δ into Q;
      $\Gamma = \Gamma \setminus \Delta$;
    end if
  end while
  $k = k + 1$, generate cluster $C_k = \Gamma_{old} \setminus \Gamma$;
  $\Omega = \Omega \setminus C_k$;
end while
return $C = \{C_1, C_2, \ldots, C_k\}$;

and then, the model parameters will be updated by the new mean vector, the new covariance matrix, and the mixture coefficient (M step). However, the GMM is sensitive to the initial clustering number because different initial numbers of clusters result in very different clustering results. AIC is introduced to measure the fitting superiority of the statistical model and determine the appropriate number of components of the GMM. The AIC is minus twice a penalized log-likelihood value, maximized using the maximum likelihood estimation (MLE) of the parameter vector θ. The likelihood function is denoted

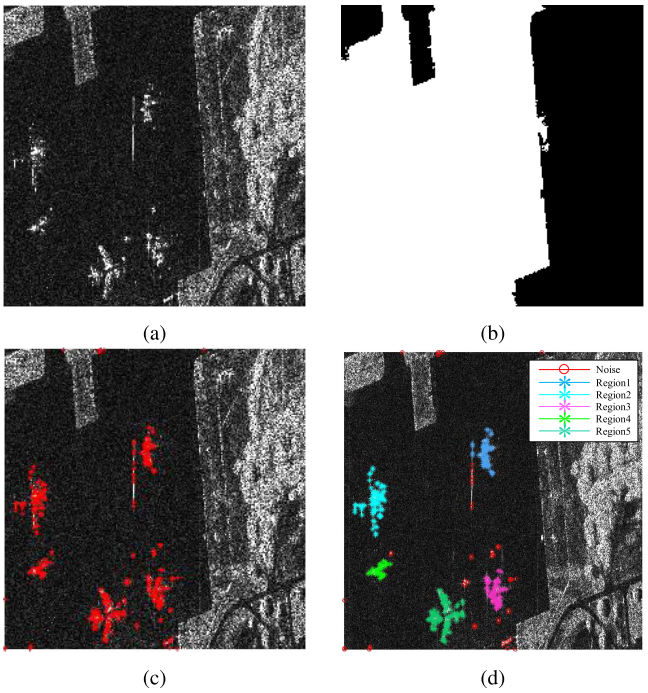

Fig. 5. Illustration of scattering information extraction stage. (a) Original SAR image. (b) Extracted airport mask. (c) Extraction results of SSPs. (d) Clustering results by DBSCAN algorithm.

as follows, with $\dim(\theta)$ the length of parameter vector θ

$$\mathrm{AIC}\{f(\cdot;\theta)\} = -2\log L\left(\hat{\theta}\right) + 2\dim(\theta). \tag{6}$$

AIC selects the model that best approximates the unknown true data generating process, among the set of models under consideration, which are generated by specifying $1, 2, \ldots, 10$ components successively to fit ten GMMs in this experiment. The number of optimization iterations is set to 1000. The bigger $\log L(\hat{\theta})$, the more precise the model, while the smaller $\dim(\theta)$, the more concise the model. Thus, the AIC balances the precision and conciseness of the model. A good model has a small value of AIC. The smaller the AIC, the better the model. An example of using different numbers of components to initialize the GMM is shown in Fig. 6. The result shows that the AIC is minimum when the number of components is 9, so the initial number of clusters for this sample is set to 9. With the initialization of the AIC, the GMM is solved by the EM algorithm. Fig. 7 shows an illustration of the scattering information modeling result for an aircraft.

Fig. 8 shows the constructed GMMs of the aircraft and background targets, where it can be seen that the distribution of aircraft GMMs and background target GMMs is distinctly different. The GMMs of the aircraft have clear structure information and inerratic component distribution, whereas the background target GMMs are irregular and it is also hard to recognize the corresponding original targets. To distinguish the aircraft among the generated GMMs, the KLD is utilized to measure the similarity between the candidate GMMs Q and the reference models P that are pretrained in a labeled aircraft data set. $p(x, y)$ and $q(x, y)$ represent the corresponding

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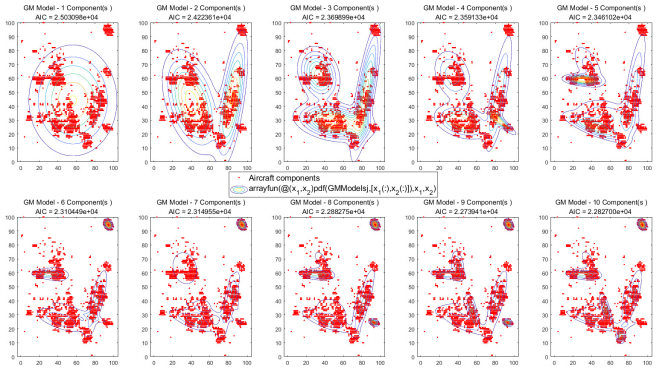

Fig. 6. GMM clustering result for an aircraft using different numbers of components to initialize and the corresponding AIC.

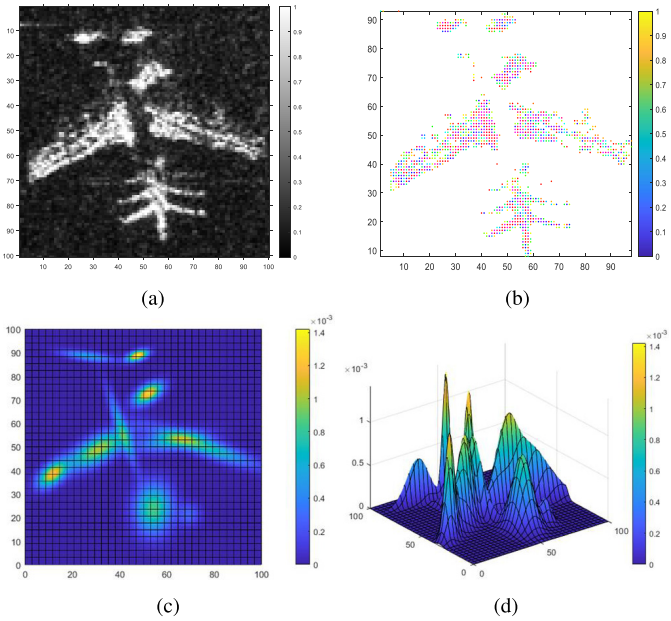

Fig. 7. Illustration of scattering information modeling result. (a) Example of an aircraft in SAR image. (b) Selected samples of the aircraft by image preprocessing in which pixel values represent the frequency. (c) 2-D illustration of constructed GMM. (d) 3-D illustration of constructed GMM.

probability density function, respectively

$$D_{\mathrm{KL}} = \mathrm{KL}(P\|Q) = \int_1^m \int_1^n p(x,y) \log \frac{p(x,y)}{q(x,y)} d_x d_y. \quad (7)$$

After the GMM modeling and KLD discrimination, the scattering regions are enhanced through Γ processing which is defined in (8), where I_{input} represents the image of the corresponding scattering region modeled and I_{CFAR} is the processed image of I_{input} by gamma distribution-based CFAR. Two-channel-enhanced processing is selected experimentally. The pseudocode of the proposed scattering region modeling and discrimination algorithm is shown in Algorithm 2

$$\Gamma\left(I_{\mathrm{input}}\right) = \begin{cases} \mathrm{cat}\left(3, I_{\mathrm{input}}, I_{\mathrm{CFAR}}, I_{\mathrm{CFAR}}\right), & D_{\mathrm{KL}} \le \text{threshold} \\ \mathrm{cat}\left(3, I_{\mathrm{input}}, I_{\mathrm{input}}, I_{\mathrm{input}}\right), & D_{\mathrm{KL}} > \text{threshold}. \end{cases} \quad (8)$$

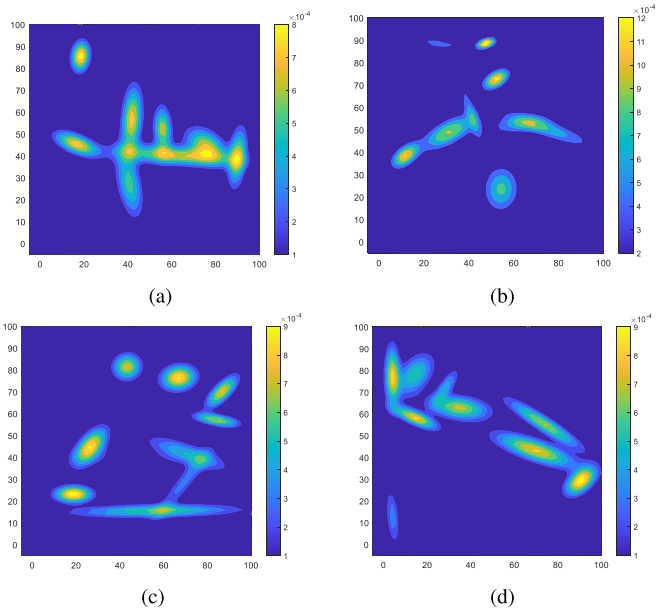

Fig. 8. Construction and visualization results of different targets. (a) Illustration of an aircraft GMM. (b) Illustration of another aircraft GMM. (c) Illustration of a background target GMM. (d) Illustration of another background target GMM.

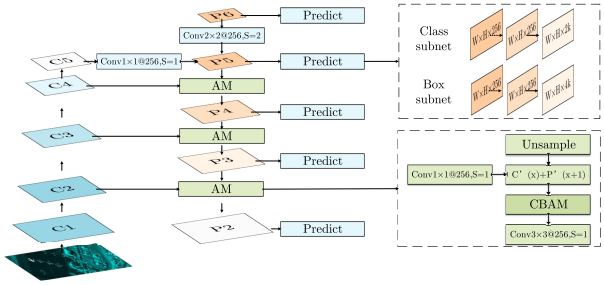

Fig. 9. Network structure of the proposed APN.

D. Attention Pyramid Network

Considering the discreteness and variability of aircraft in SAR images, APN composed of AM and FPN is proposed. Besides, the FL is introduced to cope with the unbalanced distribution of samples and improve the classification ability for hard samples. The structure of the APN is shown in Fig. 9.

1) Feature Pyramid Network Structure: In practical applications, it is quite common for large aircraft and small aircraft to coexist in SAR scenarios, and there is often quite a large difference between their scales. Large aircrafts generally have a pilot length of 40∼70 m and a wingspan of 40∼50 m, whereas small aircrafts generally have a pilot length of less than 20 m and a wingspan of less than 15 m. What is more, given the discreteness of aircraft targets in SAR images, the difference in pixel scale of large aircraft and small aircraft is considerable. In other words, for small aircraft, observation redundancy is prone to occur at a scale appropriate for observing large aircraft. This situation often affects the detection performance of small aircraft, especially in the case of dense target arrangement. On the contrary,

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Algorithm 2 Scattering Region Discrimination

Input:

Generated regions set $C = \{C_1, C_2, \ldots, C_m\}$;
Set threshold for D_{KL};

Output:

Scattering regions set $R = \{R_1, R_2, \ldots, R_m\}$;

Initialize scattering regions set $R = \emptyset$;
for $t = 1, 2, \ldots, m$ **do**
  Image preprocessing;
  Generate $X_t = \{x_1, x_2, \ldots, x_I\}$;
  Initialize Value of AIC set $V_t = \emptyset$;
  for $k = 1, 2, \ldots, 10$ **do**
    Initialize parameters of GMM $\{(\alpha_i, \mu_i, \Sigma_i)|1 \le i \le k\}$;
    repeat
      for $j = 1, 2, \ldots, I$ **do**
        Calculate posterior probability for x_j: $\gamma_{ji} = \dfrac{\alpha_i \cdot p(x_j|\mu_i, \Sigma_i)}{\sum_{l=1}^{k} \alpha_l \cdot p(x_j|\mu_l, \Sigma_l)} (1 \le i \le k)$;
      end for
      for $i = 1, 2, \ldots, k$ **do**
        Calculate new mean vector: $\mu_i' = \dfrac{\sum_{j=1}^{m} \gamma_{ji} x_j}{\sum_{j=1}^{m} \gamma_{ji}}$;
        Calculate new covariance matrix: $\Sigma_i' = \dfrac{\sum_{j=1}^{m} \gamma_{ji}(x_j - \mu_i')(x_j - \mu_i')^{\mathrm{T}}}{\sum_{j=1}^{m} \gamma_{ji}}$;
        Calculate new mixture coefficient $\alpha_i' = \dfrac{\sum_{j=1}^{m} \gamma_{ji}}{m}$;
      end for
      Update parameters $\{(\alpha_i, \mu_i, \Sigma_i)|1 \le i \le k\}$ with $\{(\alpha_i', \mu_i', \Sigma_i')|1 \le i \le k\}$;
    until satisfy stop condition
    Calculate $Value_k = AIC(\{(\alpha_i, \mu_i, \Sigma_i)|1 \le i \le k\})$;
    Update $V_t = V_t \cup \{Value_k\}$;
  end for
  Set $Q_t = \{(\alpha_i, \mu_i, \Sigma_i)|1 \le i \le K\}$ with $Value_k = \min(V_t)$, $K \in [1, 10]$;
  Calculate D_{KL} for Q_t with equation(7);
  Calculate $R_t = \Gamma(C_t)$ with equation(8);
  Update $R = R \cup \{R_t\}$;
end for
return $R = \{R_1, R_2, \ldots, R_m\}$;

when small aircraft can be observed properly, incomplete detection of large aircraft is prone to occur due to scale differences and the discreteness characteristics of aircraft in SAR images. Small aircrafts generally require high-resolution information, whereas large aircrafts are more suitable to be detected with high-semantic information. On the one hand, if a large aircraft is detected in shallow layers, it is easy to take only a small piece of the aircraft for prediction. On the other hand, when detecting small aircraft in deeper layers, too much redundant information will be contained for prediction. The former is called incomplete detection, whereas the latter is called observation redundancy. In order to cope with the observation redundancy or incomplete detection caused by the scale difference and discreteness, the FPN structure is introduced. The FPN fuses the high-resolution information of shallow features and the high-semantic information of deep features and makes predictions at different feature layers, which helps with the precise positioning of small aircraft with an appropriate scale for large aircraft observation.

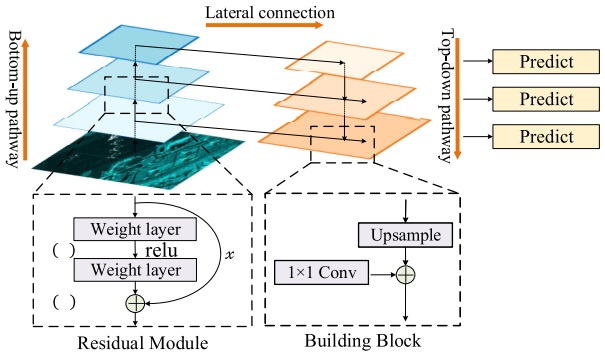

Fig. 10. Framework of FPN illustrating the building block composed of a bottom–up pathway, top–down pathway, and lateral connection.

Deep features have a powerful semantic representation ability, and therefore, most of the existing target detection algorithms only apply the top features for prediction. Although the shallow features contain less semantic information, the target location information in shallow layers is more accurate, which is very important for the target detection task. Target detection needs to predict the category and location information of a target simultaneously. Therefore, the FPN includes three major structures: the bottom–up pathway, the top–down pathway, and the lateral connection, so as to realize the feature expression of different dimensions of an image, as shown in Fig. 10.

1) *Bottom–Up Pathway:* The bottom–up pathway is the feedforward computation of the backbone ConvNet, which is assigned by ResNet101 in the proposed APN. ResNet101 is composed of five different residual modules, and the feature activations output by each stage's last residual block is used to build the FPN structure. This is reasonable since the deepest layer of each module should have the strongest features. In particular, the conv1 is not included in the pyramid due to its large memory footprint. A set of feature maps is established by the bottom–up pathway and will be enriched subsequently to create a pyramid structure.
2) *Top–Down Pathway and Lateral Connection:* The top–down pathway hallucinates higher resolution features by upsampling processing. Although the feature maps generated from higher pyramid levels are spatially coarser, they are semantically stronger. These features are then enhanced with features from the bottom–up pathway via lateral connections. Each lateral connection merges feature maps of the same spatial size from the bottom–up pathway and the top–down pathway. The bottom–up feature map is of lower level semantics, but its activations are more accurately localized as it was subsampled fewer times. This process iterates until the lowest level with the highest resolution of the feature map is generated. In particular, a 3×3 convolutional

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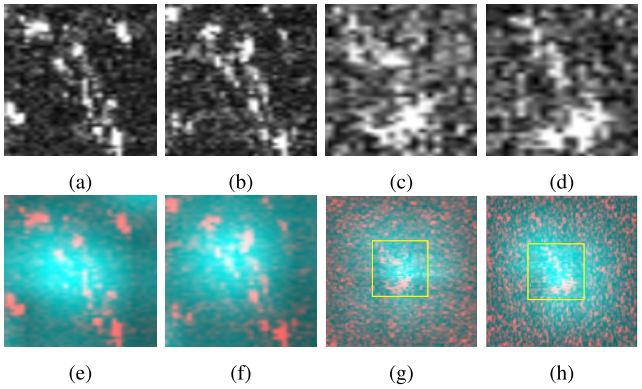

Fig. 11. Illustration of different scale aircraft feature mapping results. The size of the image slice is 128 × 128 pixels and the box mark for small aircraft is 46 × 46 pixels. (a) and (b) Examples of large aircraft in SAR imagery. (c) and (d) Examples of small aircraft in SAR imagery. (e) and (f) Corresponding feature mapping results for large aircraft. (g) and (h) Corresponding feature mapping results for small aircraft.

layer is appended to each fused feature map to generate the final feature map to reduce the aliasing effect of upsampling.

2) Attention Module Structure: In the detection of aircraft in SAR images, it is hoped that the network can pay more attention to the extraction of aircraft target features and avoid background interference. Compared with the method of increasing the depth, width, and cardinality of the network, CBAM has very low computational overhead, so a CBAM-based AM is proposed, which improves the feature extraction ability and makes the network focus more on important aircraft features while suppressing unnecessary features. It is considered that the different scales of aircraft should have different significant features; for example, for a large aircraft, the focus will be more on the positional relationship among the aircraft components, whereas for small aircraft, the focus should be more on the overall target (as shown in Fig. 11). The AM is attached behind each fusion feature layer in order to infer the attention mapping adaptive for different scales.

The AM includes two parts: the channel AM and spatial AM, which are placed in a sequential arrangement, as shown in Fig. 12. The overall attention process can be summarized as

$$F' = M_c(F) \otimes F \tag{9}$$

$$F'' = M_s\left(F'\right) \otimes F' \tag{10}$$

where $\otimes$ denotes the elementwise multiplication where the attention values are broadcast accordingly: channel attention values are broadcast along the spatial dimension, and vice versa. Given an intermediate feature map $F \in \mathbb{R}^{C\times H\times W}$ as input, the CBAM sequentially infers a 1-D channel attention map $M_c \in \mathbb{R}^{C\times 1\times 1}$ and a 2-D spatial attention map $M_s \in \mathbb{R}^{1\times H\times W}$. F'' is the final refined feature. A detailed description of each AM component is given as follows.

1) *Channel AM:* The channel AM is used to exploit the interchannel relationship of features and focuses on "what" is meaningful for given feature maps. Both average- and max-pooled features are utilized simultaneously for the max-pooling operation, which is not adopted in the SE module but is considered to gather another important clue about distinctive objects to infer finer channelwise attention. The channel AM M_c is computed as

$$\begin{aligned} M_c(F) &= \sigma(\mathrm{MLP}(\mathrm{AvgPool}(F)) + \mathrm{MLP}(\mathrm{MaxPool}(F))) \\ &= \sigma\left(W_1\left(W_0\left(F^c_{\mathrm{avg}}\right)\right) + W_1\left(W_0\left(F^c_{\mathrm{max}}\right)\right)\right) \end{aligned} \tag{11}$$

where F^c_{avg} and F^c_{max} denote the average-pooled features and max-pooled features, respectively, σ represents the sigmoid activation function, and W_0 and W_1, which denote the multilayer perceptron weights, are shared for both pooling features.

2) *Spatial AM:* The spatial AM is used to exploit the interspatial relationship of features and focuses on "where" is meaningful for given feature maps. Similarly, both average- and max-pooled features are utilized for inferring the attention mapping and are concatenated along the channel axis. On the concatenated feature descriptor, a convolution layer is applied to generate a spatial attention map $M_s(F)$, which encodes where to emphasizes or suppress. The detailed operation is described in (12), where $f^{7\times 7}$ represents a convolutional operation with a filter size of 7×7

$$\begin{aligned} M_s(F) &= \sigma\left(f^{7\times 7}\left(\left[\mathrm{AvgPool}(F); \mathrm{MaxPool}(F)\right]\right)\right) \\ &= \sigma\left(f^{7\times 7}\left(\left[F^s_{\mathrm{avg}}; F^s_{\mathrm{max}}\right]\right)\right). \end{aligned} \tag{12}$$

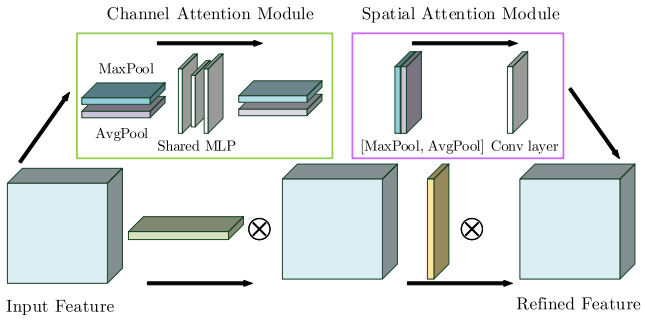

Fig. 12. Structure of CBAM.

3) Prediction Stage: The network for aircraft detection in SAR imagery needs to predict the category and position information of the target simultaneously, where the CE loss and smooth-L1 loss are generally utilized for predicting the labels and positions, respectively. However, in large-scene SAR imagery, the aircraft is prone to appear as a large number of discrete highlighted scattering points due to its discrete and variable characteristics, which are easily confused with the complex background objects. CE loss, as illustrated in (13), cannot obtain a satisfactory classification performance in this case, where the distribution of positive and negative samples is unbalanced and a large number of difficult samples exist. Therefore, the FL is designed to address the mentioned challenges

$$\mathrm{CE}(p, y) = \begin{cases} -\log(p), & \text{if } y = 1 \\ -\log(1-p), & \text{otherwise} \end{cases} \tag{13}$$

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where $y \in \pm 1$ specifies the ground-truth label and $p \in [0, 1]$ represents the estimated probability for the class with label $y = 1$. For notational convenience, CE can be defined with p_t as follows:

$$p_t = \begin{cases} p, & \text{if } y = 1 \\ 1 - p, & \text{otherwise .} \end{cases} \tag{14}$$

Furthermore, the FL $\mathrm{FL}(p_t)$ is designed with a weighting factor α_t and a tunable focusing parameter γ as

$$\mathrm{FL}(p_t) = -\alpha_t (1 - p_t)^{\gamma} \log(p_t). \tag{15}$$

There are two properties of the FL function. When an example is misclassified and p_t is small, the modulating factor $(1-p_t)^{\gamma}$ is near to 1 and the loss is unaffected. As p_t is near 1, the factor goes to 0 and the loss for well-classified examples is downweighted; the focusing parameter γ smoothly adjusts the rate at which easy examples are downweighted. When $\gamma = 0$, the FL is equivalent to CE, and as γ is increased, the effect of the modulating factor is likewise increased ($\alpha_t = 0.25$ and $\gamma = 2$ are selected in our experiments).

IV. Experiments and Analyses

A. Data Set Description

An experimental data set composed of GF-3 SAR images and TerraSAR-X images is established to illustrate the efficiency and robustness of the proposed method. The ground truth of the experimental data is manually labeled by comparing it with the optical image, which is considered reliable in this article. Examples of the experimental data set are shown in Fig. 13. Other detailed information of the data set is described in the following.

1) GF-3 Aircraft Detection Data Set (GF3-ADD): GF-3 is a C-band spaceborne SAR that can work in 12 different modes, including SL, UFC, FSI, and SS. A total of 69 C-band HH polarization images with 1 m $\times$ 1 m resolution in SL mode are used in this experiment. Images in GF3-ADD cover various complex scenarios, in which it is hard to distinguish the targets among background noise. In addition, aircrafts adjacent to buildings are common in this data set. Therefore, GF3-ADD is established to verify the performance of the proposed algorithm in a complex background.

2) TerraSAR-X Aircraft Detection Data Set (T-ADD): TerraSAR-X is an X-band SAR that can work in Spotlight, Stripmap, and ScanSAR modes. A total of three HH polarization images with 1 m $\times$ 1 m resolution are selected in this experiment to test the performance of the proposed algorithm with targets in a densely arranged situation.

As shown in Fig. 14, the GF3-ADD data set contains a variety of airport scenes, so the size distribution of aircraft is relatively uniform. The pilot length is mostly in a range of 10–60 m, with a wingspan of 5–40 m. In the T-ADD data set, the target distribution of aircraft with partial size is denser. The experimental data set, including GF3-ADD and T-ADD, is cut into slices with 1000 $\times$ 1000 pixels after AD and the training set, as well as the test set, are divided with a ratio of 8:2. The detailed information for the experimental data set is given in Table I.

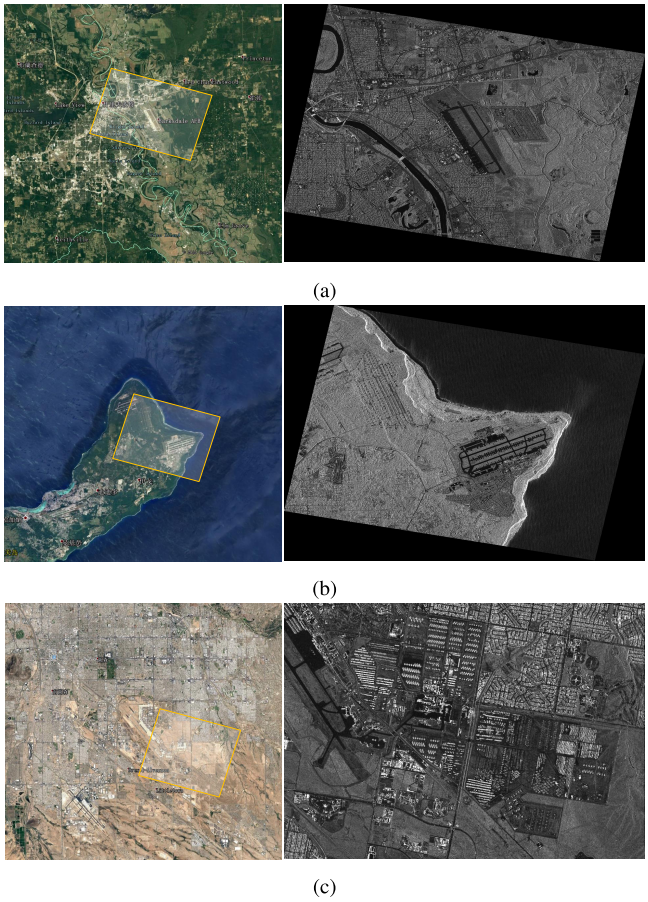

Fig. 13. Illustration of the experimental data. (a) Example from GF3-ADD. (b) Another example of data from GF3-ADD. (c) Example from T-ADD.

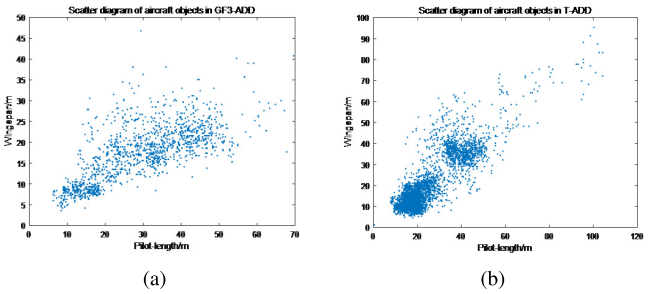

Fig. 14. Scatter diagram of aircrafts in the experimental data sets. (a) Scatter diagram of aircrafts in GF3-ADD. (b) Scatter diagram of aircrafts in T-ADD.

TABLE I
Information of Experimental Data Set

Data set	Training	Test	Total
GF3-ADD	528	132	660
T-ADD	426	106	532
Total	954	238	1192

B. Evaluation Metrics

In order to evaluate the performance of the proposed algorithm quantitatively, some evaluation metrics are employed, including the false alarm rate, precision, recall, average precision (AP), F1 score, and speed. These metrics are calculated

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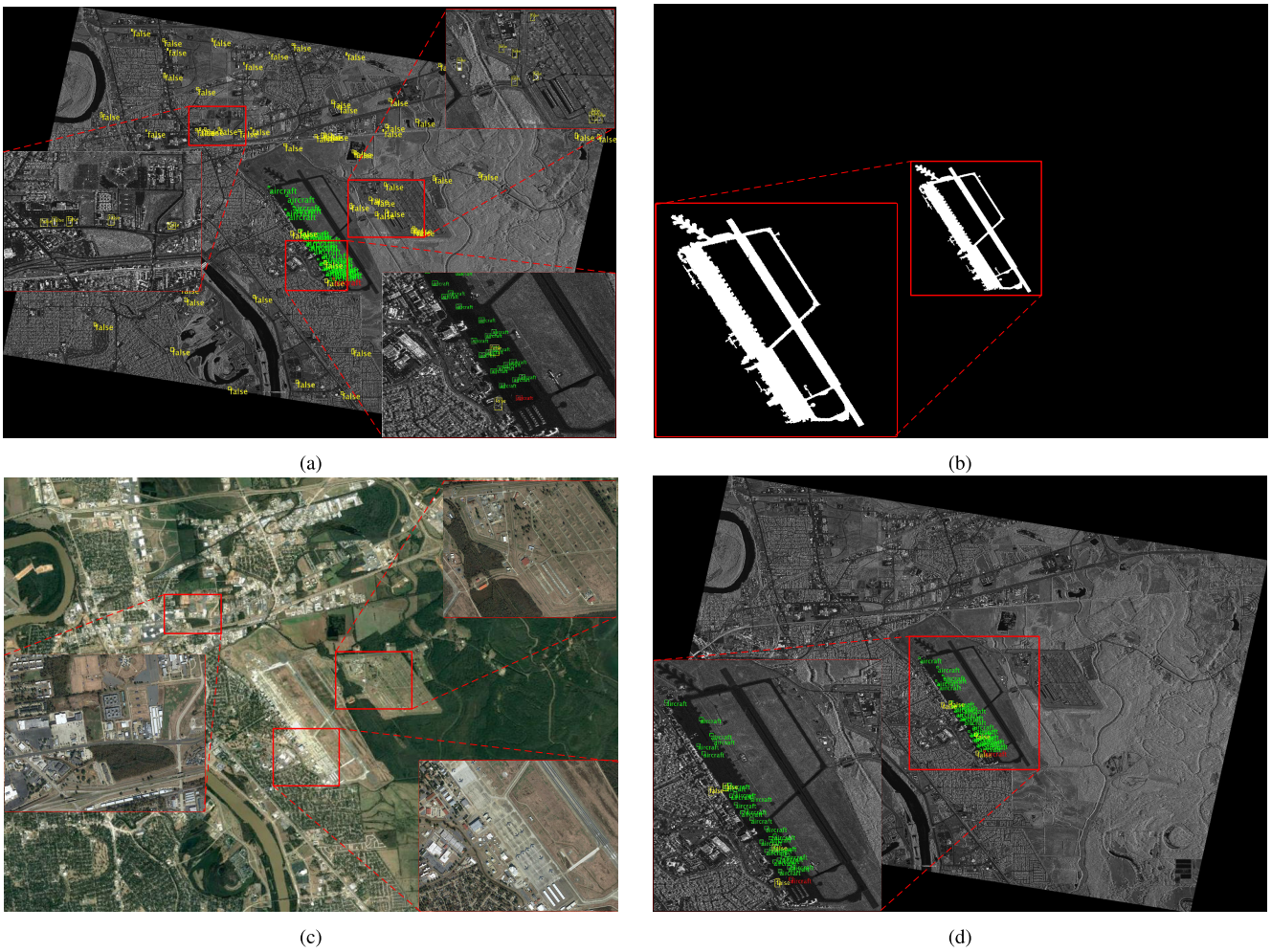

Fig. 15. Illustration of the effectiveness of AD algorithm. (a) Aircraft detection result without AD in a Gaofen-3 image. (b) Airport mask extracted by the proposed AD algorithm. (c) Corresponding optical image for the same scene. (d) Aircraft detection result with AD in a Gaofen-3 image.

by five well-established values measured in the experiments: true positive (TP), false positive (FP), true negative (TN), false negative (FN), and running time for each scene. In this experiment, TP and FP represent the correctly detected aircraft and the false alarms, respectively. TN represents the correct undetected background targets, whereas FN represents the undetected aircraft. All outputs are arranged in descending order of confidence. For a given confidence, a set of detection results whose confidences are higher than the threshold will be selected. Generally, the threshold is set as 0.5. The definitions of these assessments are as follows.

1) *False Alarm Rate:* It represents the proportion of FP in the overall detected results and reflects the credibility of the algorithm. The lower the false alarm rate is, the better the detector is performing

$$\text{False alarm rate} = \frac{\text{FP}}{\text{TP} + \text{FP}}. \tag{16}$$

2) *Precision:* It is the ratio of correctly detected targets in the overall detected results and represents the correctness of the detected objects. The definition is as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}. \tag{17}$$

3) *Recall:* It is defined as the proportion of correctly detected targets in the overall true targets. Recall and precision are a pair of contradictory measures that are widely used for evaluating the performance of detectors. Generally, when the recall is higher, the precision is relatively low and vice versa

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{18}$$

4) *AP:* For a given task or class, the precision/recall curve is computed from a ranked output. The AP summarizes the shape of the precision/recall curve at different confidence level from 0 to 1 and can be expressed as

$$\text{AP}_d = \int_0^1 P_d(R_d)\text{d}(R_d) \tag{19}$$

where d is the intersection over union (IoU) threshold to determine whether a detected bounding box is truly positive or false positive. True positive with an IoU is higher than $d\%$ between the ground truth and detection. P_d and R_d represent precision and recall, respectively. d is set to 50 in the evaluation.

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TABLE II
EFFECTIVENESS OF THE AIRCRAFT DETECTION ALGORITHM

Algorithm	False alarm rate	Precision	Recall	FPS (scene)
Without AD	15.1%	84.9%	93.7%	0.042
AD	7.9%	92.1%	93.7%	0.083

5) *F1 Score:* $F1$ is defined based on the harmonic mean of precision and recall and can evaluate the overall performance of detection algorithms. $F1$ is computed as

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{20}$$

6) *Frames per Second (FPS):* It represents the number of images (frames) processed within 1 s. The higher FPS is, the faster speed a method achieves

$$\text{FPS} = \frac{1}{T_{\text{per-img}}} \tag{21}$$

where $T_{\text{per-img}}$ refers to the inference time cost by a method when processing a single image.

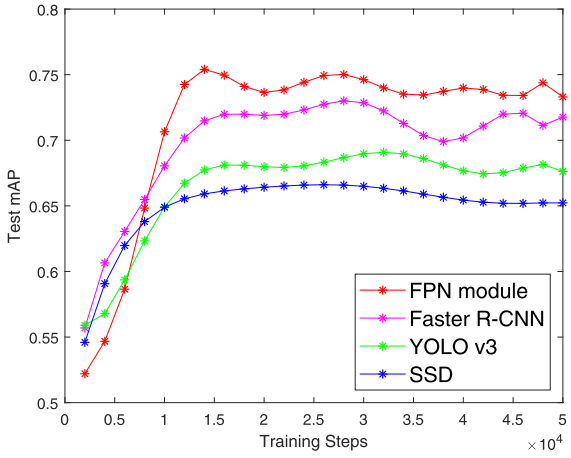
C. Parameters Setting

It has two stages.

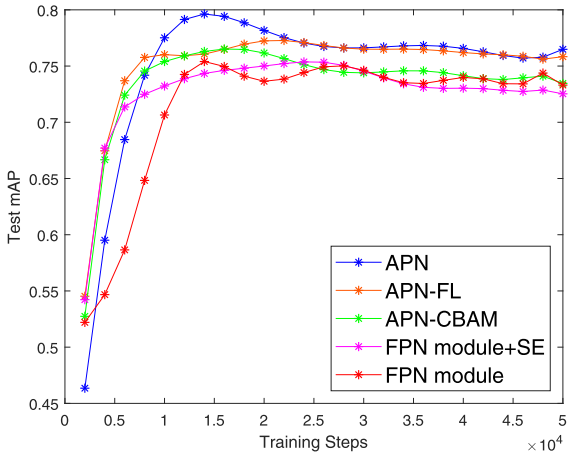
1) *Parameters in the SIE Stage:* In the DBSCAN step, we set $\epsilon = 20$ for T-ADD and $\epsilon = 30$ for GF3-ADD with MinPets $= 5$ experimentally. In addition, when judging the similarity between the mixed Gaussian models of different targets, considering the small difference between the aircraft target classes and the large difference between the aircraft target and the background clutter, the threshold value θ is set to 20 empirically.
2) *Parameters in the APN Stage:* The input size for the APN is 1000×1000 pixels, and the basic anchor size for P2, P3, P4, P5, and P6 is set to 32, 64, 128, 256, and 512, respectively. In order to compare the performance of different detection algorithms, the learning rate is set to $1e^{-}5$ with a weight decay $= 1e^{-}4$ as well as momentum $= 0.9$, and the number of training steps is set to 50 000.

D. Experimental Results and Analysis

1) Effectiveness of AD: An accurate AD algorithm is beneficial to reduce computation as well as avoid a multitude of false alarms. The effectiveness of AD is given in Table II. The size of the test image is 14 670 $\times$ 12 285 pixels. The aircraft detection method used here is the proposed algorithm with/without AD processing. As can be seen from the table, the false alarm rate decreases obviously, and precision increases as well. Because of the AD process, a smaller searching area shortens the detection time by almost a half. An illustration of the effectiveness of the proposed AD algorithm is shown in Fig. 15.



(a)



(b)

Fig. 16. Effectiveness of the proposed APN. (a) Comparison with state-of-the-art detection algorithms: effectiveness of FPN module. (b) Effectiveness of AM and FL.

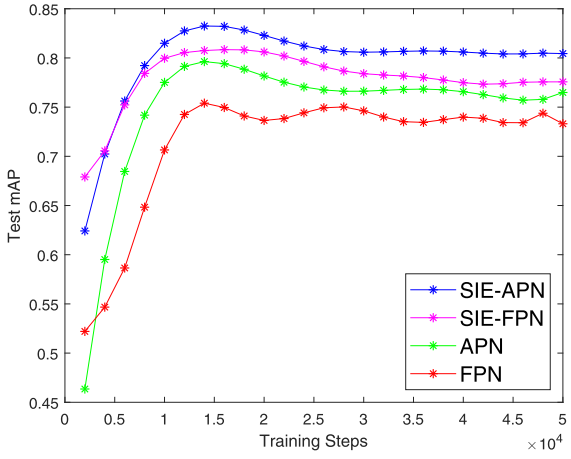


Fig. 17. Effectiveness of the proposed SIE processing.

2) Effectiveness of APN: Based on the structure of the APN, experiments were designed to verify the effectiveness from two aspects: the FPN and AM. Comparisons were also made between the FPN module and typical detection algorithms,

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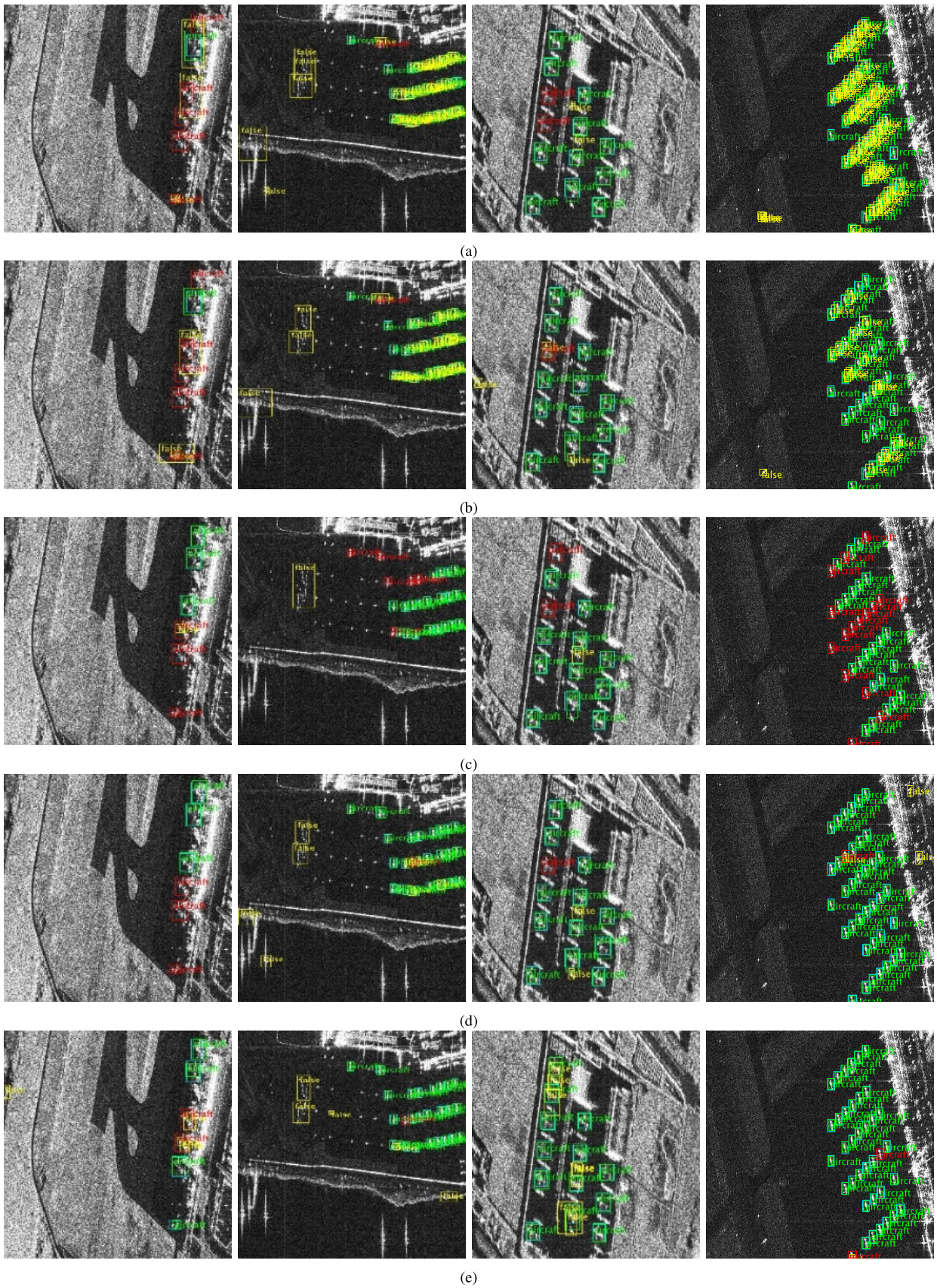

Fig. 18. Comparison results of each algorithm mentioned in several test samples of GF3-ADD: The cyan boxes, yellow boxes, red boxes, and green boxes represent ground truth, false alarms, missed targets, and detected targets, respectively. (a) SSD. (b) YOLO v3. (c) Faster R-CNN. (d) FPN module. (e) FPN + SE.

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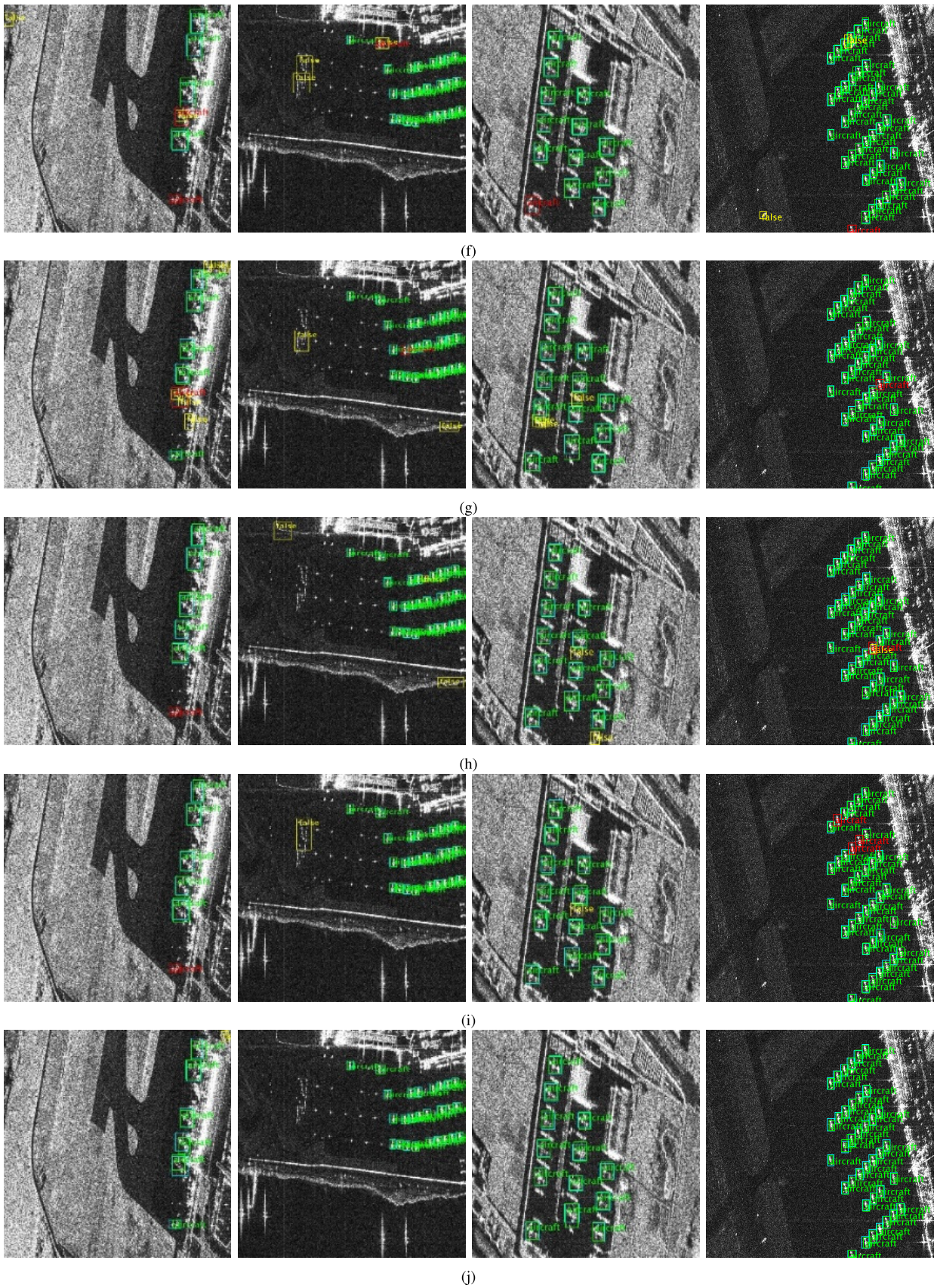

Fig. 18. *(Continued.)* Comparison results of each algorithm mentioned in several test samples of GF3-ADD: The cyan boxes, yellow boxes, red boxes, and green boxes represent ground truth, false alarms, missed targets, and detected targets, respectively. (f) APN-CBAM. (g) APN-FL. (h) APN. (i) SIE-FPN. (j) SIE-APN.

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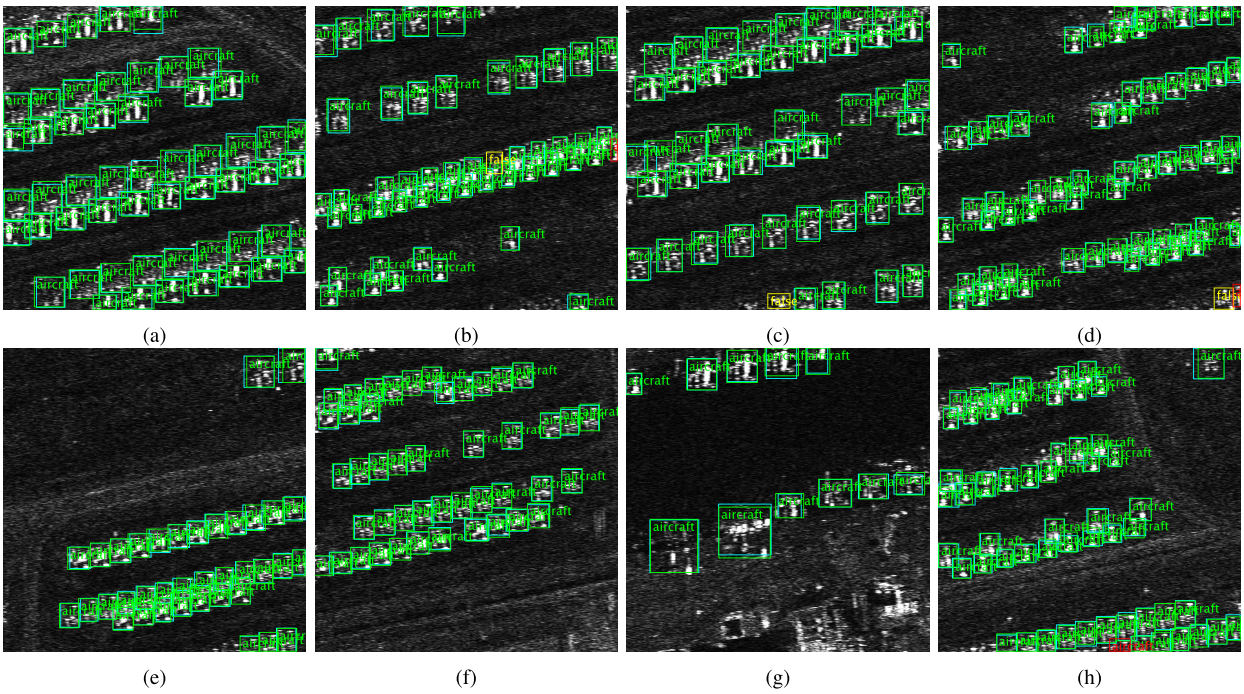

Fig. 19. SIE-APN detection results in several test samples of T-ADD. The cyan boxes, yellow boxes, red boxes, and green boxes represent ground truth, false alarms, missed targets, and detected targets, respectively. (a) Region 1. (b) Region 2. (c) Region 3. (d) Region 4. (e) Region 5. (f) Region 6. (g) Region 7. (h) Region 8.

including SSD, YOLO v3, and Faster R-CNN. The results demonstrated that the FPN module performed better too. As shown in Fig. 16(a), the test AP of the FPN is, respectively, 8.79%, 6.31%, and 2.39% higher than SSD, YOLO v3, and Faster R-CNN. Furthermore, by analyzing the experimental results, it is found that the FPN module can effectively improve the network's detection performance for small targets.

To illustrate the effectiveness of the AM, experiments were designed on different attention mechanisms involving the CBAM and SE [53], as well as the loss function FL. As shown in Fig. 16(b), the results demonstrated that the introduction of a proper AM can improve the performance. It is worth noting that the test AP decreases when the SE module is added to the network. This may be because too much information is lost in the single Avg-Pool stage. However, the connection mode combining Avg-Pool with Max-Pool is better than that in single pooling. Besides, using FL instead of standard CE loss, the overall detection performance can be improved to some extent. Compared with the FPN module, when only the CBAM module (APN-FL) or only FL was utilized (APN-CBAM), the test AP increased by 1.87% and 1.12%, respectively. Encouragingly, when both of them were added, the test AP increased by 4.23%.

3) Effectiveness of SIE: Furthermore, an extended experiment was conducted to demonstrate the effectiveness of SIE processing. The experimental results are shown in Fig. 17 and demonstrate that the detection performance can be significantly improved by enhancing the scattering information for aircraft candidates in the original image. The network converges faster and the test results are more stable. Compared with FPN, the test AP for the proposed algorithm APN with SIE (SIE-APN) increases by 7.85%. In addition, the test AP for SIE-APN increases by 3.62% compared with the APN. It is also worth mentioning that after adding SIE processing to the FPN module (SIE-FPN), the performance also became better, which illustrates the effectiveness and superiority of SIE again.

TABLE III
COMPARISON WITH OTHER ALGORITHMS

Algorithm	Recall	Precision	F1	AP	FPS (slice)
SSD	0.7311	0.6137	0.6673	0.6661	128
YOLO v3	0.7256	0.6720	0.6978	0.6909	62
Faster R-CNN	0.7253	0.7431	0.7341	0.7301	12
FPN	0.7699	0.7396	0.7544	0.7540	17
FPN+SE	0.7504	0.7528	0.7516	0.7537	16
APN-CBAM	0.7516	0.7713	0.7613	0.7652	18
APN-FL	0.7862	0.7946	0.7904	0.7727	17
APN	0.8182	0.7958	0.8068	0.7963	17
SIE-FPN	0.8212	0.7973	0.8091	0.8084	13
Ours	**0.8658**	**0.8276**	**0.8463**	**0.8325**	13

Detailed comparisons of the different algorithms and the proposed SIE-APN with different components are given in Table III. To better illustrate the performance of each algorithm, four typical SAR images in the GF3-ADD data set were selected for testing to evaluate the detection performance for aircraft in complex backgrounds, and eight typical SAR images in the T-ADD data set were selected to evaluate the detection performance for aircraft in dense arrangement situations. The results are given in Figs. 18 and 19 and correspond to each line in Table III, where yellow and red boxes denote false alarms and missed targets, respectively. At first glance, these two colors cannot be found in Fig. 18(j).

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SIE-APN means that our proposed method obtained the best performance, as listed in Table III.

In summary, the proposed SIE-APN algorithm can significantly improve aircraft detection performance and effectively suppress false alarms in both complex background and dense arrangement situations.

V. Conclusion

A novel method for aircraft detection in high-resolution SAR images based on the combination of SIE and an APN is proposed in this article. In the SIE stage, the SSPs in the scene are first extracted by a Harris–Laplace detector, and then, the scattering region candidates are generated by DBSCAN. In addition, a GMM is introduced to model these candidates with AIC initialization and KLD is utilized to measure the similarity between the pretrained reference model and the test model. In this stage, scattering regions containing aircraft can be enhanced by the scattering information, which means that two channels of the scene image in this region are replaced with the image processed by the Gamma distribution-based CFAR. Through this stage, the network can focus more on target feature extraction, avoiding the interference of background clutter. In the APN stage, a novel structure, which is composed of FPN module and AM, is proposed to cope with the discreteness and variability. Moreover, the FL is introduced to relieve the influence caused by the unbalanced distribution of positive and negative samples as well as the influence of hard samples. The test AP of experiments conducted on the GF3-ADD and T-ADD data sets reaches 83.25%, which demonstrates the robustness and superiority of the proposed algorithm in both complex background and target dense arrangement situations.

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